

A Comprehensive Investigation on Leveraging Generative AI and Large Language Models in the Healthcare Domain

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Abstract—The launch and adoption of Generative Pre-Trained Transformers (GPT) based AI has been very rapid compared to any other technological adoption seen in the past. In these times of the rapid evolution and advancement of Large Language Models (LLMs) and Generative Artificial Intelligence (GenAI), measuring and reviewing the impact it has made on the industry, especially the healthcare domain, becomes very important. Since the healthcare sector is one of the early adopters of novel technologies in the industry, the penetration of LLMs and GenAI has been predominant in the sector. In this paper, we are understanding and analysing the impacts of this novel technology in the healthcare sector, categorised under different domains such as Education, Medical Information Management, Telemedicine and Telehealth, Health Record Management, Health Data Research, Mental Health Care and Personal Health Planning. Additionally, the use of LLM and GenAI in the healthcare domain raises some critical ethical concerns as well. This paper tries to investigate the potential applications and ethical challenges of GenAI and Large Language Models (LLMs) within the healthcare domain.

Keywords—Generative Pre-Trained Transformers, Large Language Models, Generative Artificial Intelligence, Medical Information Management, Telemedicine and Telehealth, Health Record Management, Health Data Research, Mental Health Care, Personal Health Planning

I. INTRODUCTION

LLM based Artificial Intelligence (AI) models have been in the forefront of the AI domain.

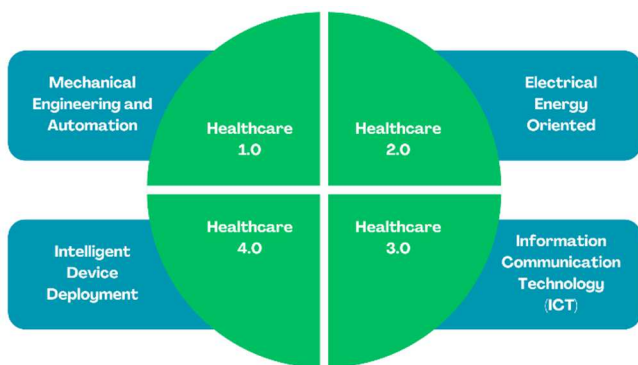


Fig.1. Generations in Healthcare and Technology Advancements

The popularisation and mass adoption of these models only happened after the launch of Chat based GPT models,

which gave a preview of what it looks like to interact with an AI. Since its popularity, the adoption in industrial and commercial applications are revolutionising many areas within. The integration of LLM [1] based AI models are not limited to the industry sector, but has also influenced the healthcare domain, by improving efficiency, ease of data analysis and interpretation, and for providing personalised recommendations according to the requirements of the patient. It is important to measure the impact made by these LLM models in the healthcare industry, to understand and track the ethical practices that have to be followed. In this paper, there has been an effort to measure and study the impacts of GenAI in the healthcare industry, and identify the ethical concerns that need to be addressed as well as the possible solutions applicable for the same.

II. OVERVIEW

A. Advancement of AI

AI has made significant advancements in technology, surpassing any other technological revolutions. The rapid development and adoption of AI are attributed to its ease of training models by providing more accurate data. This is similar to the transistor evolution, where the transition from 50K individual transistors to 30 billion transistors in microprocessors took at least a year [2]. With abundant data, newer AI versions can be upgraded within months or weeks, demonstrating the rapid development and adoption of AI.

The adoption of new AI in the industrial space will face a barrier of proper due diligence, but once completed, it can be easily integrated into existing systems with a lower cost of entry compared to past technological advancements. However, the medical sector has been the exception, with only certain exceptions like AI in image processing and detection of abnormal cells [3]. The wide adoption of AI has only been possible after the introduction of LLM-based GenAIs, which have significantly reduced the initial capital and transition time required for proper implementation.

B. What is GenAI?

Generative AI or GenAI is a trained AI model that uses algorithms and deep learning techniques to analyse existing data, find relevant patterns and use them to generate new customised information or data, according to the prompt or input given by the user. This can be new sentence formation, image generation or even a new code. GenAI [4] uses a

variety of algorithms, especially deep learning, to identify patterns in existing data and then creates completely original, creative material, in contradiction to traditional AI, which examines pre-existing data. Deep learning models, a class of artificial neural networks (ANNs) trained on vast datasets, are vital for GenAI. These models leverage the intricate relationships it discovers in the data to generate new results. It is anticipated that GenAI will advance in capability and provide ever more imaginative and accurate results as research continues. It will be important for many industries, but responsible development and taking ethical concerns into account are important.

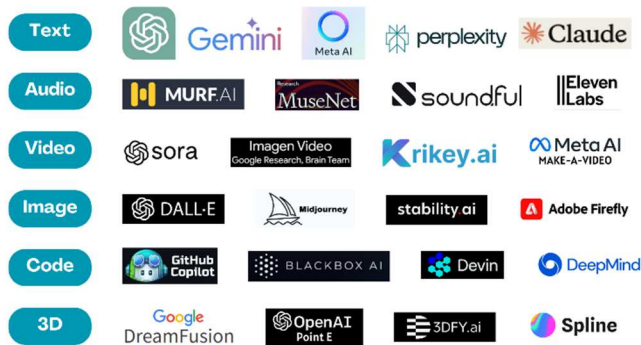


Fig.2. Technologies in GenAI and LLM Across Verticals

C. What are LLMs?

Large Language Models (LLMs) are AI algorithms that analyse and process large amounts of text data to generate unique human language texts. They have the capacity to answer prompts, summarise, paraphrase, or translate texts. Models like ChatGPT and Gemini [5] have been used in various industries, such as healthcare. Integrating LLMs into software can create personalised software for diets and lifestyles. However, ethical concerns and reliability of these models must be evaluated under proper guidelines and regulations. The application of LLMs in healthcare is limitless.

D. LLM vs GenAI: A Brief Comparison

Although Large Language Models and Generative AI [6] feels like both are similar to each other and can be used interchangeably, the same is not true. While LLM models deal with understanding and manipulating existing information, GenAI focuses on creating entirely new data through creative generation. For the sake of comparison with an analogy, LLM models act as a librarian or a scholar, which has access to a large amount of data. On the other hand, GenAI is like a creative artist, which can generate new information. Both of these are subsets of AI with distinct functionalities and purposes. GenAI and LLMs are two growing technologies that have the ability to completely reinvent a wide range of sectors in the industry. As they advance, the distinction between human creativity and machine-generated content becomes difficult to distinguish which leads to increasingly complex applications and adoption of these models

III. EVOLUTION OF LLM & GENAI

Large Language Models and Generative AI [7] have become some of the most popular and widely adapted AI technologies, which has put the spotlight on these technologies. The fact about these models of AI is that it can learn continuously and adapt according to the users' input

prompts. Since the public release of ChatGPT and Gemini (formerly Bard) the advancement and evolution of these technologies have been very rapid, as the number of users and the data available is humongous. The version upgrades and generation changes have been faster than ever compared to any other technological revolution that happened in the past. The same goes with the wide adoption of the technology, across the spectrum of industries and academic research.

IV. APPLICATIONS IN HEALTH CARE

The healthcare industry is embracing technology to improve health outcomes. LLM and GenAI [8] are promising technologies for driving innovation and improving the system. LLM models analyse medical data for diagnosis, suggest treatment plans, and personalise care. GenAI automates patient report generation, personalises treatments, and creates synthetic medical data for research. These technologies can support clinical administration and provide professional education tools for patients and healthcare providers. The integration of AI in healthcare can enhance patient care, facilitate medical research, and alleviate workload on healthcare professionals.

I. AVAILABLE LLM MODELS FOR HEALTHCARE

S. No.	AI Model	Application	Ref.
1	Med-PaLM	Med-PaLM (Medical Pre-Trained Language Model) offers excellent responses to medical queries with the capacity to comprehend medical jargon, decipher intricate medical ideas, and produce pertinent comments or insights.	[9]
2	BioGPT	Pre-trained transformer language model that is domain-specifically generative and designed for producing and mining biomedical texts.	[10]
3	IBM Watson for Oncology	GenAI based model created to aid oncologists in selecting the best course of treatment for cancer patients.	[11]
4	NVIDIA Clara	A collection of AI-powered tools and frameworks created primarily for healthcare applications, specifically trained to comprehend and interpret content written in medical natural language.	[12][13]
5	DeepHealth LLM	The model attempts to close the knowledge gap between unstructured clinical data and insights that healthcare professionals may use.	[14][15]
6	BioBERT	BioBERT (Bidirectional Encoder Representations from Transformers for Biomedical Text Mining) is a LLM model for biomedical text mining and natural language processing (NLP) activities in healthcare.	[16]
7	Med7 LLM	An LLM based clinical text data extraction tool trained on a wide variety of clinical text data such as electronic health records (EHRs), clinical notes, and medical literature.	[17]

V. MEDICAL EDUCATION

GenAI can provide personalized information for medical students, doctors, nurses, staff, and patients. LLM models have made significant advancements, with Med-PaLM 2 and GPT-4 reaching 86.5% and 86.1% accuracy respectively. These models can be tailored to specific needs and preferences. LLMs can generate trustworthy, explainable outputs for improving medical education, guiding learners in writing papers with strong internal consistency and a logical framework [20]. Integrating them into clinical decision-making procedures may be a feasible endeavour in the future.

VI. HEALTH INFORMATION MANAGEMENT AND DOCTOR'S PERSONAL ASSISTANT

LLMs can act as health information assistants, providing specific patient information based on patient history. They can integrate with digital health records, acting as doctors' assistants. GenAI [22] can revolutionise clinical workflows by assisting medical professionals in interpreting data from various sources, including genomics, imaging records, medical histories, and laboratory findings. This text-based assistant can analyse complete patient data, enabling better patient assessment and saving time compared to individual document analysis.

VII. TELEMEDICINE AND TELEHEALTH

During the COVID-19 pandemic, telemedicine and telehealth have seen significant technological advancements [23]. The introduction of LLM and GenAI-based solutions has paved the way for innovation and evolution in these areas. These technologies can enhance patient-provider interaction, automate tasks, and personalize healthcare experiences. Applications include enhanced virtual assistants, personalized treatment recommendations, real-time language translation, and automated report generation [24]. However, concerns such as data privacy, security, limited physical examination capabilities, regulation, ethical considerations, and the digital divide remain.

VIII. HEALTH RECORD MANAGEMENT

Digitalisation [25] of patient medical records has improved data management and accessibility in hospitals worldwide. However, challenges arise in data segregation and analysis. LLM models can help by understanding large amounts of data and providing relevant output. Integrating these models in health record management benefits various hospital departments, including physicians, patients, and administration. The data can also be used for medical research, removing unnecessary information and keeping patient data private. GenAI can help design a prompt-based data output system that contributes significantly to scientific research by addressing the challenges of analyzing this vast amount of patient data.

IX. HEALTH DATA RESEARCH AND GENERATE SYNTHETIC DATA

Scholars and researchers face challenges in extracting adequate medical data, particularly vital patient information, due to ethical and privacy concerns and hospital policies. GenAI [26] has developed a concept of synthetic medical data generation to address these issues. This data can be used to augment training data, create diverse datasets for AI models, and generate artificial environments to simulate real-life patient conditions. This technology can be integrated with

AR/XR-based realities for improved accuracy and efficiency in research and medical training [27].

A. GenAI for Drug Development

GenAI models can accelerate drug development [28] and molecular structure development in healthcare by generating novel molecular structures with desired properties through multiple iterations and considering multiple parameters simultaneously. This technology is expected to accelerate clinical and scientific developments at an unprecedented pace, making it more efficient and time-saving, eliminating the need for multiple clinical trials and laboratory experiments [29].

X. MENTAL HEALTH CARE

Mental healthcare [30] has become increasingly important, especially after the COVID-19 pandemic, as people become more aware of the importance of mental health. However, in developed countries like the US, there is a 350:1 ratio of available professionals, with limitations such as limited accessibility and personalisation. Factors affecting people daily include anxiety, frustration, and depression. GenAI can improve the quality of life by providing personalised treatment plans and therapy based on individual needs. GenAI can analyse vast amounts of data, identify patterns indicating mental health issues, and identify those in need of urgent care. Generative adversarial networks and hierarchical attention networks can diagnose depression using multi-modal data like text and audio physiological signals. By leveraging GenAI's [31] power in mental healthcare, it can benefit people in terms of accessibility, effectiveness, and personalisation.

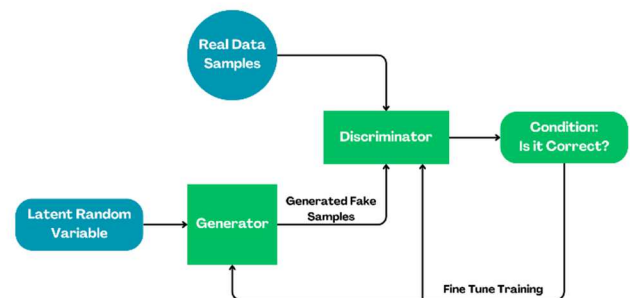


Fig.3. Generative Adversarial Network (GAN) Model Flowchart

XI. PERSONAL HEALTH PLANNING

LLM models [32] have been increasingly used in personalised health planners, acting as at-home dieticians or gym trainers. These models can design dedicated meal plans or workout routines, utilising user data. For instance, a knowledge-infused LLM-powered conversational health agent (CHA) for diabetic patients is trained to incorporate American Diabetes Association [33] dietary guidelines and Nutritionix information. The model uses GPT-4, a GenAI model, to provide specialised meal choices and suggest risks related to the user's food consumption. ChatDiet, a new LLM-powered architecture, is designed for chatbots to provide personalised meal recommendations focusing on nutrition [34]. It uses an orchestrator, population, and personal models to process and retrieve relevant data, resulting in dynamic, personalised, and tailored meal recommendations.

XII. MEDICAL CODING AND EHRs

Electronic Health Records (EHRs) [35] are digital versions of patient files, offering benefits such as accessibility, improved communication, decision-making, and reduced administrative burden. The integration of Long-Long Memory (LLM) models in EHR management can provide relevant information in an optimized order. Medical coding [36] transforms healthcare diagnoses, procedures, services, and equipment into standardized alphanumeric codes, useful for billing, insurance, and data utilization. LLM and GenAI models can help medical coders automate code prediction, identify missing information, resolve coding queries, and improve coding accuracy. This increases efficiency, accuracy, reduces costs, and frees up time for coders. The integration of LLM and GenAI models can revolutionize EHR management and medical coding.

XIII. MEDICAL TRIAGE

Medical triage [38] is a crucial process of quickly assessing and prioritising injured or sick patients to determine their treatment order. This is especially important in situations with limited resources, such as doctors or infrastructure. Physicians often have to make difficult decisions based on various parameters and variables. The introduction of LLM-based models can help analyse patient parameters and prioritise them in order, saving time and avoiding bias. This can save millions of lives, reduce stress on healthcare providers, and provide better results. LLM models' ability to understand and analyse data provides the best possible outcomes in these situations.

XIV. DISEASE PROGRESSION MODELLING

The healthcare sector is focusing on personalised treatment [39], but predicting disease progression in individual patients remains a challenge. Traditional methods rely on statistical analysis, which doesn't provide a comprehensive picture. LLM and GenAI [40] can help with disease progression modelling by analysing large amounts of medical data, identifying patterns, and extracting insights from unstructured data. GenAI can simulate patient progression on a virtual patient, generating personalised disease progression models and designing interventions. This can inform treatment decisions, slow disease progression, and improve patient outcomes. However, challenges include data availability and quality, model accuracy, and ethical considerations. Overall, these technologies offer promising opportunities for personalised medicine and early intervention.

XV. AI HALLUCINATION

AI tools like ChatGPT [41] have gained widespread acceptance and usage, with over 1.5 billion people globally using them. However, AI hallucination, the generation of misleading or incorrect information, can be difficult to identify due to factors like insufficient training data, incorrect assumptions, or biases. To mitigate AI hallucinations, high-quality training data, well-curated datasets, continuous monitoring, and human oversight are essential. These factors help ensure the accuracy and effectiveness of AI models, especially in high-stakes situations.

XVI. MEDICAL ETHICS

- a) Advancements in healthcare technology are revolutionising the medical sector. However, the

heavy reliability of LLM and GenAI models [43] on existing biased data increases the risk of misinformation. In clinical settings, eliminating biases is crucial for personalised medication and patient confidentiality. GenAI models should be trained on data free of gender, racial, social, religious, or cultural biases.

- b) Th GenAI and LLM models are inevitable in healthcare due to their transformative potential. However, their integration should be cautiously regulated by regulatory bodies like FDA, EU [44], and WHO. Despite challenges in predicting their evolution, attempts have been made to create regulatory laws for these AI models.
- c) The EU AI Act [45], the first Western regulation of Artificial Intelligence for industrial applications, mandates the review of GenAI models before commercial release. Critics argue it affects developers like OpenAI, Google, and Meta, and has raised concerns about human rights activists and open-source communities.
- d) The implementation of GenAI and LLM models in healthcare systems requires a regulatory body to oversee and regulate AI use, ensuring patient safety, medical ethics, and privacy, while working alongside existing global regulatory bodies.

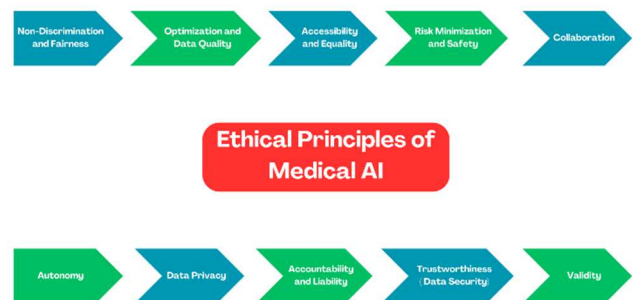


Fig.4. Ethical Principles of AI in Healthcare

- e) AI deployment in healthcare necessitates robust legal and ethical frameworks to address privacy, data utilization, biases, and AI hallucinations. Regulatory bodies can mitigate error by continuously monitoring advancements and innovations in the medical sector.
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XVII. THE PRODUCTIVITY PARADOX

The productivity paradox [46] in the healthcare industry is evident in the introduction of Electronic Health Records (EHRs), which has not led to expected efficiency improvements due to hidden costs, data overload, information fatigue, data entry issues, and increased administrative burden. Privacy policy regulations in healthcare affect data sharing, and the complex implementation of GenAI [47] in the sector is difficult for non-EHR companies. Cluttered data and the "fail first and

iterate" approach of AI models also contribute to the paradox. The stakes in healthcare are too high to tolerate flaws, making it crucial to address these issues.

XVIII. CONCLUSION

The rapid evolution of AI models, particularly LLM and GenAI models, is crucial in the healthcare sector. These models are used in various verticals such as medical image analysis, chatbots for personalized healthcare support, healthcare operations and resource management, optimization of clinical trials, drug discovery enhancement, medical simulation generation, training, and analysis.

The paper highlights the technology investments available in the healthcare landscape, including Med-PaLM, a medical query-based assistant, and IBM Watson, a clinical decision-making assistant specialized in oncology. EHR data-based LLM models simplify the heavy load on hospital administration by managing and analyzing patient data, reducing stress on healthcare professionals and providing informed decision-making.

GenAI has contributed significantly to research by generating simulations of human movements, understanding prompts, and providing generative results. It also supports research-based applications and training LLM-based models. Disease Progression Modelling helps understand the current scenario of patients and potential disease progression, enhancing prevention possibilities.

Combining LLM and GenAI models leads to more advanced applications in healthcare, such as medical coding and health record management. The former automates medical coding processes and the latter acts as a doctor's assistant, analyzing medical history and disease conditions for individual patients.

However, there are limitations to integrating AI models into healthcare systems, such as AI hallucination due to biased data training or limited information availability. Proper AI laws and regulatory frameworks are needed to ensure ethical and reliable use of AI models. As AI continues to advance, there are significant opportunities for improvement in the medical technology domain.

XIX. FUTURE PROSPECTS

AI research and development are rapidly progressing due to vast data and modern computer efficiency. This has streamlined model integration, but there is still room for improvement in accuracy and precision. Many models give false or hallucinated results, creating a productivity paradox due to slow adoption of these technologies. AI models are being enhanced for better clinical decision-support [48], making them highly effective and reliable for medical applications. Future prospects include enhanced clinical decision support with more accurate and personalised results, enabling healthcare service providers to act as intelligent assistants in diagnosis, treatment planning, and patient management. A shared environment for information exchange between practitioners from different disciplines can save time and improve patient health. Disease prediction and

prevention [49] can be improved by enhancing LLM and LAM models, along with advanced ML models. Natural Language Processing (NLP) can help bridge the communication gap between patients and healthcare service providers, making patient information more accessible and simplifying healthcare professionals' work [50]. AI has also been beneficial in streamlining administrative operations, such as patient data management and data allocation. GenAI and LLM models can analyse large amounts of data, identify parameters, and provide results for decision-making situations. However, proper upgrades are needed for real-world scenarios. Additionally, AI should be responsible for mitigating and proposing proper regulatory frameworks for the use of different AI models [51].

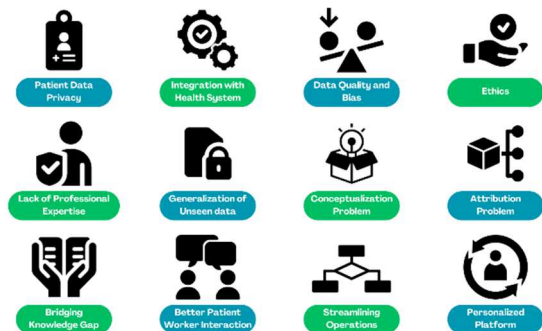


Fig.5. Limitations & Future Works for Integration of LLM and GenAI in Healthcare

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